13

Lecture 13

Serving and PDN Gateways (S-GW and P-GW)

دروازه‌های خدمت‌دهنده و (S-GW PDN) و P-GW)

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

DIFFICULTY
Intermediate

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

PREREQUISITES
Lectures 9 and 12; IP addressing, UDP, tunnelling

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 4
6 Mathematical models . . . 6
7 Network architecture and diagrams . . . 7
8 Step-by-step procedures . . . 8
9 Worked examples . . . 9
10 Technology comparison . . . 9
11 Key takeaways . . . 10
12 Exercises . . . 10
13 Further reading . . . 12

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-13-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Distinguish** the functions of the serving gateway and the PDN gateway, and **explain** why the user plane has two anchors.
- **LO2** **Describe** GTP-U tunnelling and **compute** its overhead and its effect on the MTU.
- **LO3** **Trace** downlink delivery to an idle UE and the path switch after a handover.
- **LO4** **Model** APN-AMBR policing with a token bucket and **size** gateway buffers and packet rates.
- **LO5** **Explain** control/user-plane separation (CUPS) with PFCP rules, and **compare** home-routed and local-breakout roaming.

2 Prerequisite knowledge

This lecture uses the bearer model of Lecture 9 (default and dedicated bearers, TFTs, QCI, AMBR) and the MME procedures of Lecture 12 (attach, paging, service request). From networking you need IPv4/IPv6 addressing, UDP and the idea of tunnelling: carrying one packet as the payload of another.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
S-GW	دروازهٔ خدمت‌دهنده	User-plane anchor for mobility between eNBs; buffers for idle UEs.
P-GW	دروازهٔ PDN	Anchor towards a packet data network; allocates the UE's IP address.
GTP-U	پروتکل تونل‌زنی GPRS، کاربر صفحهٔ	UDP-based tunnelling protocol for user packets [4].
TEID	شناسهٔ نقطهٔ پایانی تونل	32-bit identifier of a tunnel at the receiving node.
PCEF	تابع اعمال سیاست و صورت‌حساب	Enforcement point of PCC rules in the P-GW.
DDN	اعلان دادهٔ فروسو	S-GW's request to the MME to page an idle UE.
End marker	نشانگر پایان	GTP-U packet that marks the last packet on the old path after a switch.
Token bucket	سطل نشانه	Policing model with rate r and depth b.
CUPS	جداسازی صفحهٔ کنترل و کاربر	Split of each gateway into a control and a user-plane function [2].
PFCP	پروتکل کنترل پیش‌برد بسته	Protocol by which the control plane installs packet rules [5].

3 Introduction and motivation

Every byte that an LTE subscriber sends or receives crosses two gateways. The *serving gateway* (S-GW) faces the radio network: it stays fixed while the UE moves between eNBs and holds packets while the UE sleeps. The *PDN gateway* (P-GW) faces the Internet or the IMS: it gives the UE its IP address and enforces the operator's policy and charging. The P-GW is the UE's point of presence in the IP world, so it must not change during a session; the S-GW is a regional anchor and may change with mobility [3].

Gateways are where the core meets traffic volume. An MME is dimensioned in procedures per second; a gateway is dimensioned in bits and packets per second, and its design is dominated by forwarding performance, buffering and per-packet rule processing. This lecture follows a packet through the gateways, quantifies the cost of tunnelling and policing, and ends with the separation of control and user planes that leads to the 5G UPF.

4 Fundamental concepts

Table 1. Functions of the S-GW and the P-GW [3].

S-GW	P-GW
Local mobility anchor for inter-eNB handover; sends end markers	UE IP address allocation (IPv4, IPv6 prefix)
Downlink buffering in ECM-IDLE; triggers paging via DDN	Policy enforcement (PCEF): gating, QoS, APN-AMBR
Mobility anchor towards 2G/3G (S4, S12)	Downlink packet filtering with TFTs to select the bearer
Lawful interception	Lawful interception, deep packet inspection
Per-bearer accounting (e.g. for roaming settlement)	Online (Gy) and offline (Gz) charging
DSCP marking on the transport network	DSCP marking, NAT and firewall functions (operator choice)

4.1 Division of labour

A UE has exactly one S-GW at a time, which serves all its PDN connections; it may have several P-GWs, one per PDN connection (for example one for the Internet APN and one for IMS). Operators often deploy a *combined S/P-GW* for non-roaming traffic, in which S5 becomes internal.

4.2 Tunnels instead of routes

The UE's IP address belongs to the P-GW's address pool. If routers between eNB and P-GW forwarded the UE's packets by their IP header, every handover would require routing updates. Instead each bearer is carried in GTP-U tunnels: S1-U between eNB and S-GW, S5/S8 between S-GW and P-GW (Figure 1). A handover only changes the tunnel endpoint (address and TEID) at the S-GW; the UE's address and the P-GW never change. Each bearer has its own pair of tunnels in each direction, identified by the receiver's TEID.

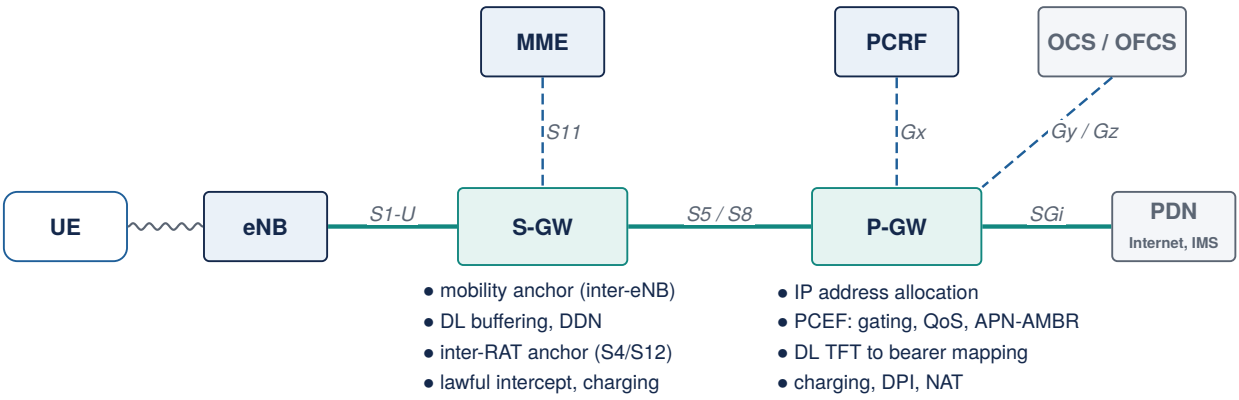


Figure 1. S-GW and P-GW on the user-plane path with their main functions.

5 Detailed technical explanation

5.1 GTP-U

GTP-U runs over UDP port 2152 [4]. Its mandatory header is 8 bytes: flags (version 1, protocol type, and the E, S and PN bits that announce 4 optional bytes for an extension header, a sequence number and an N-PDU number), the message type (255 for G-PDU, i.e. user data), the length of what follows the mandatory header, and the TEID (Figure 2). Signalling messages of GTP-U itself include Echo Request/Response for path supervision, Error Indication (the receiver has no context for a TEID) and End Marker (type 254). The inner packet (T-PDU) is not modified: the gateways see and police the UE's own IP header.

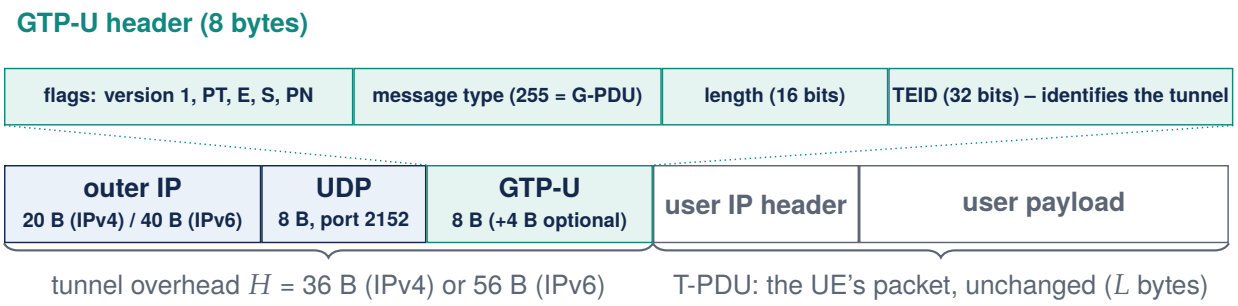


Figure 2. GTP-U encapsulation of a user packet and the mandatory GTP-U header.

5.2 IP address allocation

At attach (or PDN connection set-up) the UE requests PDN type IPv4, IPv6 or IPv4v6. For IPv4 the P-GW returns an address in the Create Session Response, which reaches the UE in NAS; for IPv6 it allocates a /64 prefix and announces it in Router Advertisements over the default bearer, from which the UE builds its address. Private IPv4 addresses with carrier-grade NAT at or behind the P-GW are common, which is one motivation for IPv6-only APNs.

5.3 Policy enforcement

The PCEF in the P-GW applies the PCC rules received over Gx (Lecture 10): it maps downlink packets to bearers with DL TFTs, blocks or allows service data flows (gating), polices GBR/MBR per bearer and APN-AMBR across the non-GBR bearers of a PDN connection (Section 6.3), counts volume and time per rating group for charging, and marks packets with DSCP values derived from the QCI so that the transport network can prioritise them. UE-AMBR is enforced by the eNB.

5.4 Idle mode and handover

When a UE goes to ECM-IDLE, the eNB releases its context and the MME asks the S-GW to release the S1-U tunnels; the S5/S8 part stays. A downlink packet then finds no S1-U tunnel: the S-GW buffers it and sends DDN to the MME, which pages the UE (Figure 4). After an X2 handover the target eNB asks the MME for a path switch; the MME sends Modify Bearer Request to the S-GW, which switches the downlink to the target eNB and sends one or more *end marker* packets on the old path, so that the source eNB knows when forwarding of in-flight packets is complete and the target can deliver packets in order.

5.5 Control/user-plane separation (CUPS)

Release 14 split each gateway into a control function (SGW-C, PGW-C) and one or more user-plane functions (SGW-U, PGW-U), connected by the Sx interfaces [2]. The control function keeps S11, S5-C, Gx and Gy; the user-plane function only forwards packets according to rules installed with PFCP [5]:

- **PDR** (packet detection rule): which packets (TEID, UE address, SDF filter) the rule applies to;
- **FAR** (forwarding action rule): forward, drop, buffer or duplicate, and where to;
- **QER** (QoS enforcement rule): gating and rate limits;
- **URR** (usage reporting rule): what to count and when to report;
- **BAR** (buffering action rule): how to buffer for idle UEs.

User-plane functions can then be placed close to the radio network (for low latency and local traffic) and scaled independently of the control plane. The 5G UPF and its N4 interface reuse this model (Lecture 23).

6 Mathematical models

6.1 Tunnelling overhead and MTU

Model. A user packet of L bytes is carried with tunnel overhead H (outer IP + UDP + GTP-U): $H = 20 + 8 + 8 = 36$ bytes with IPv4 transport, $H = 56$ bytes with IPv6. The transport efficiency and the largest user packet that fits a transport MTU M without fragmentation are

$$\eta = \frac{L}{L+H}, \qquad L_{\max} = M - H. \tag{1}$$

Example. $L = 1400$, IPv4 transport: $\eta = 97.5\,\%$. With $M = 1500$, $L_{\max} = 1464$ bytes (1444 with IPv6 transport), which is why operators configure the UE's MTU below 1500 or enlarge the transport MTU.

Interpretation. Overhead matters for small packets: for voice it is a large fraction of the packet (Example 1). Header compression (ROHC) on the radio does not help in the core, where the full headers and the tunnel are carried.

6.2 Downlink buffering

Model. A source sends at rate R to an idle UE; until the S1-U tunnel is restored after a paging and service-request delay D, the S-GW must buffer

$$B = R\,D. \tag{2}$$

Example. $R = 2\,\text{Mbit/s}$, $D = 1.5\,\text{s}$: $B = 3\,\text{Mbit} = 375\,\text{kB}$. In practice TCP sources send only a few packets before waiting for acknowledgements, so actual buffers are much smaller, and gateways limit the buffer per bearer and drop beyond it.

6.3 Policing with a token bucket

Model. A token bucket with rate r and depth b admits a traffic stream if, for every interval of length T, the data sent $A(T)$ satisfy

$$A(T) \le b + r\,T. \tag{3}$$

A source with peak rate $R > r$ starting with a full bucket can send at its peak for $T^{*} = b/(R-r)$.

Example. APN-AMBR $r = 10\,\text{Mbit/s}$, $b = 8\,\text{Mbit}$ (1 MB): in 3 s at most $8 + 30 = 38\,\text{Mbit}$ (4.75 MB) pass; a 40 Mbit/s source can burst for $8/30 = 0.27\,\text{s}$ before being limited to 10 Mbit/s (Figure 3).

Interpretation. The rate fixes the long-term share; the depth determines how bursty traffic may be. Too small a bucket hurts TCP (losses at every burst), too large a bucket lets short bursts exceed the subscribed rate noticeably.

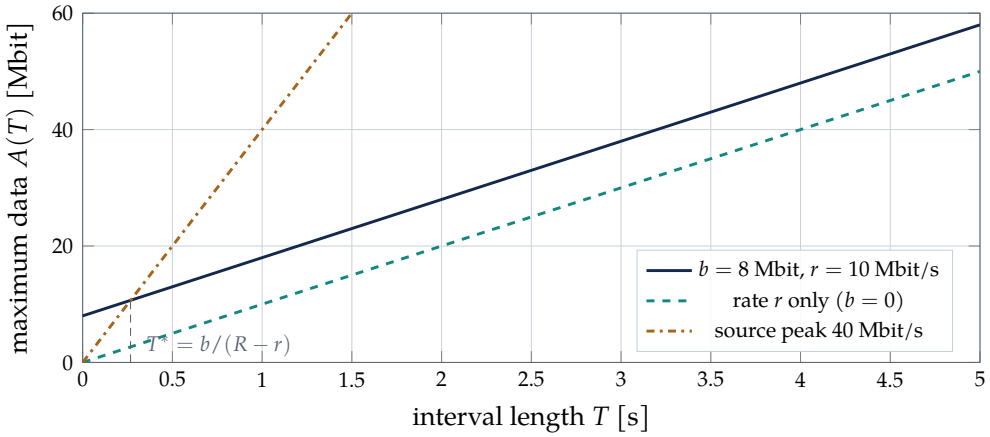


Figure 3. Token-bucket envelope $b + rT$ (Eq. (3)) and a peak-rate source.

6.4 Packet rate

Model. Forwarding cost is per packet, not per byte. A gateway carrying throughput C with mean packet size $\bar{L}$ handles

$$P = \frac{C}{8\,\bar{L}}\ \text{packets/s}. \tag{4}$$

Example. $C = 40\,\text{Gbit/s}$, $\bar{L} = 800\,\text{bytes}$: $P = 6.25\,\text{Mpps}$ – each packet must be classified against the PDRs of its session, counted and encapsulated within about 160 ns on average per packet for a single processing pipeline, which is why user planes use kernel-bypass packet I/O and hardware offload.

7 Network architecture and diagrams

In a non-roaming network the S-GW and P-GW are usually co-located in regional data centres; with CUPS, user-plane functions move towards the edge while the control functions stay central. For roaming, the visited network provides the S-GW; the P-GW is either in the home network (*home-routed*, over S8, the usual case for data) or in the visited network (*local breakout*, used e.g. for

IMS voice in some agreements). Home routing keeps charging and policy in the home network but lengthens the path (Exercise 5).

8 Step-by-step procedures

Procedure 1 • Downlink data for an idle UE

1. A downlink packet reaches the P-GW, which maps it to a bearer with the DL TFT and forwards it over S5 to the S-GW.
2. The S-GW has no S1-U tunnel for the bearer: it buffers the packet and sends Downlink Data Notification to the MME (further packets are buffered without a new DDN).
3. The MME pages the UE in its TAI list; the UE answers with a Service Request.
4. The MME sets up the context in the eNB (Initial Context Setup), learns the eNB's TEID and sends Modify Bearer Request to the S-GW.
5. The S-GW forwards the buffered packets to the eNB, followed by new traffic.

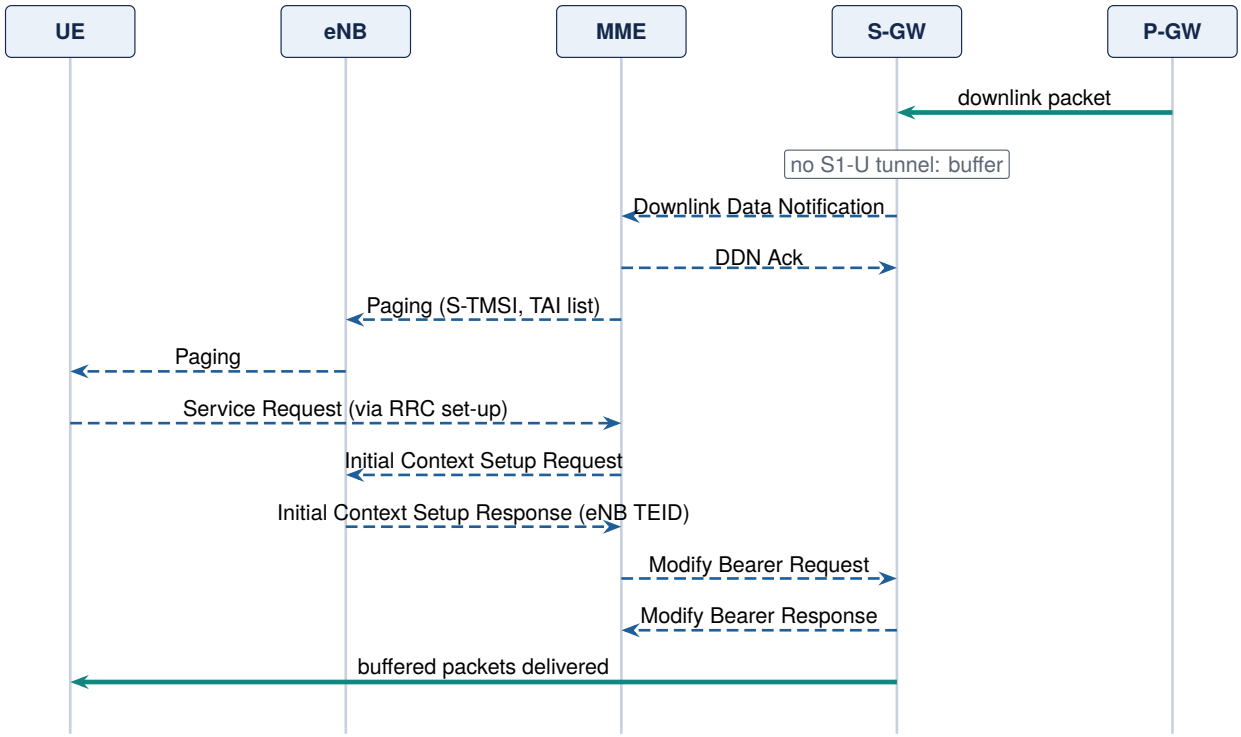


Figure 4. Network-triggered service request with S-GW buffering (Procedure 1).

Procedure 2 • Path switch after an X2 handover

1. During the handover the source eNB forwards in-flight downlink packets to the target eNB over X2.
2. The target eNB sends Path Switch Request to the MME with its downlink TEIDs.
3. The MME sends Modify Bearer Request to the S-GW, which switches the downlink tunnels to the target.

4. The S-GW sends end-marker packets to the source eNB on the old tunnels; the source relays them to the target, which then delivers the forwarded packets before the new ones.
5. The source eNB releases the UE context.

9 Worked examples

Worked example 1 • Overhead of a voice packet

Problem. An AMR-WB 12.65 kbit/s frame produces about 33 bytes of RTP payload every 20 ms; the UE uses IPv6. Compute the size of the packet on S1-U (IPv4 transport) and the fraction that is headers.
Solution. Inner packet: $33 + 12\,(\text{RTP}) + 8\,(\text{UDP}) + 40\,(\text{IPv6}) = 93$ bytes. On S1-U: $93 + 36 = 129$ bytes. Tunnel overhead alone is $36/129 = 28\,\%$; all headers together are $(129 - 33)/129 = 74\,\%$. At 50 packets/s the call uses $129 \times 8 \times 50 = 51.6$ kbit/s on S1-U for 12.65 kbit/s of speech.

Worked example 2 • Token-bucket admission

Problem. With $r = 10$ Mbit/s and $b = 8$ Mbit, a user downloads a 5 MB file at 40 Mbit/s from a server, starting with a full bucket. How long does the download take?
Solution. 5 MB = 40 Mbit. During the initial burst of $T^* = 0.27$ s, $40 \times 0.27 = 10.7$ Mbit pass; the remaining 29.3 Mbit pass at 10 Mbit/s in 2.93 s. Total ≈ 3.2 s, versus 1 s unpoliced and 4 s with $b = 0$.

Worked example 3 • Gateway packet rate

Problem. A P-GW serves 2×10^5 simultaneously active users with a mean of 200 kbit/s each. Compute throughput and packet rate for a mean packet of 800 bytes.
Solution. $C = 2 \times 10^5 \times 2 \times 10^5 = 40$ Gbit/s; $P = 40 \times 10^9/6400 = 6.25$ Mpps (Eq. (4)). Uplink packets (many small TCP acknowledgements) add to the packet rate while adding little throughput.

10 Technology comparison

One anchor or two?. Two anchors separate concerns: the S-GW hides radio mobility from the P-GW, and the P-GW hides the PDN from the radio side. The cost is an extra hop and extra state. 5G keeps the idea but makes it flexible: a session may have one UPF or a chain of them, chosen per session by the SMF.

11 Key takeaways

Table 2. Packet gateways across generations.

	SGSN + GGSN (2G/3G)	S-GW + P-GW (4G)	UPF (+ SMF) (5G)
Anchors	GGSN (IP), SGSN (mobility)	P-GW (IP), S-GW (mobility)	PDU-session anchor UPF, intermediate UPFs
Tunnelling	GTP-U (Gn, Iu-PS)	GTP-U (S1-U, S5/S8)	GTP-U (N3, N9)
Control of forwarding	Integrated	Integrated; CUPS from Rel-14	Always separated: PFCP over N4
QoS granularity	PDP context	EPS bearer	QoS flow (QFI)
Local breakout / edge	Rare	Possible with CUPS	Native (UL classifier, branching point)

Key takeaways

- S-GW: mobility anchor between eNBs, buffering and DDN for idle UEs, end markers.
- P-GW: IP anchor (address allocation), PCEF (TFT mapping, gating, AMBR, charging), SGi to the PDN.
- GTP-U tunnels carry bearers; overhead $H = 36$ (IPv4) or 56 (IPv6) bytes, $L_{\max} = M - H$.
- Buffering $B = RD$; policing $A(T) \leq b + rT$; forwarding cost $P = C/(8\bar{L})$ packets/s.
- CUPS splits gateways into control and user planes connected by PFCP rules (PDR, FAR, QER, URR, BAR) – the direct ancestor of the 5G UPF.

12 Exercises

Exercise 1 • Level 1 – conceptual • Which gateway?

Which gateway (S-GW or P-GW) changes in each case: (a) X2 handover between two eNBs of the same S-GW area; (b) the UE moves to a region served by another S-GW; (c) the UE opens a second PDN connection to the IMS APN?

Exercise 2 • Level 1 – conceptual • End markers

Explain what could go wrong at the target eNB after a handover if the S-GW did not send end-marker packets.

Exercise 3 • Level 2 – numerical • MTU

The transport network has an MTU of 1500 bytes and uses IPv6. What is the largest UE packet that avoids fragmentation? What MTU would the transport network need to carry 1500-byte UE packets?

Exercise 4 • Level 2 – numerical • Buffer sizing

An S-GW limits the buffer per bearer to 256 kB. For a paging and service-request delay of 1.2 s, up to which source rate is no packet dropped? If 10^4 UEs are paged simultaneously and each buffer fills, how much memory is needed?

Exercise 5 • Level 3 – analytical • Home-routed roaming

A subscriber roams 4000 km from the home network and accesses a server located in the visited country. Estimate the extra round-trip time of home routing compared with local breakout (propagation at 2×10^8 m/s, ignoring processing), and discuss why operators still prefer home routing for data.

Exercise 6 • Level 3 – analytical • Scaling a user plane

A PGW-U core forwards 1.5 Mpps. How many cores are needed for 100 Gbit/s with a mean packet of 700 bytes if cores may be loaded to at most 70 %? What does CUPS allow the operator to do when traffic doubles but the number of subscribers does not?

Solutions

Solution to Exercise 1

(a) Neither: only the S1-U endpoint (eNB) changes. (b) The S-GW changes (S-GW relocation); the P-GW and IP address stay. (c) A new P-GW (possibly) is added for the new PDN connection; the S-GW is shared.

Solution to Exercise 2

The target receives packets on two paths: forwarded over X2 from the source and directly from the S-GW. Without an end marker it cannot know when the forwarded stream has ended; delivering new packets first would reorder the flow (hurting TCP), and waiting by timer would add delay. The end marker closes the old path explicitly.

Solution to Exercise 3

$L_{\max} = 1500 - 56 = 1444$ bytes. To carry 1500-byte UE packets the transport MTU must be at least $1500 + 56 = 1556$ bytes (or 1540 with IPv4 transport); operators often run jumbo frames in the backhaul.

Solution to Exercise 4

$R \le B/D = 256 \times 8$ kbit$/1.2$ s $= 1.71$ Mbit/s. Worst-case memory $10^4 \times 256$ kB $= 2.56$ GB; in practice far less because TCP sources stall after one window and most bursts are a few packets.

Solution to Exercise 5

Home routing sends each packet visited $\rightarrow$ home $\rightarrow$ server (in the visited country) and back the same way: an extra 2×4000 km each direction, i.e. 4×4000 km$/(2 \times 10^8$ m/s$) = 80$ ms of extra RTT. Operators accept it because the home network keeps control of policy, charging, lawful interception and service access (e.g. home APNs), and because roaming settlement is

simpler; local breakout needs agreements on policy and charging in the visited network.

Solution to Exercise 6

$P = 100 \times 10^9/(8 \times 700) = 17.9$ Mpps; $17.9/1.5 = 11.9$ cores at full load, $11.9/0.7 = 17.0 \rightarrow$ 18 cores. With CUPS the operator adds user-plane capacity (more PGW-U instances or cores, possibly at new edge sites) without touching the control plane, whose load depends on subscribers and procedures, not on traffic volume.

13 Further reading

Gateway functions are specified in TS 23.401 [3], GTPv2-C in TS 29.274 [1] and GTP-U in TS 29.281 [4]; CUPS and PFCP in TS 23.214 [2] and TS 29.244 [5]; policy and charging control in TS 23.203 [6]. Olsson et al. [7] discuss gateway deployment, and Rommer et al. [8] the evolution towards the 5G user plane.

References

[1] 3GPP, *3GPP evolved packet system (EPS); evolved general packet radio service (GPRS) tunnelling protocol for control plane (GTPv2-C); stage 3*, TS 29.274, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/29274.htm.

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[8] S. Rommer, P. Hedman, M. Olsson, L. Frid, S. Sultana, and C. Mulligan, *5G Core Networks: Powering Digitalization*. Academic Press, 2020.